



AI-Driven Low-Carbon Transformation in Supply Chains: The Mediating Role of Green Dynamic Capabilities and the Moderating Role of Low-Carbon Policies

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Abstract: Low-carbon transformation of supply chains is important for climate-change mitigation, yet the mechanisms through which artificial intelligence (AI) contributes to low-carbon performance remain unclear. This study examines the associations between predictive, decision-making, and automated AI applications and low-carbon performance, considering the mediating role of green dynamic capability and the moderating role of low-carbon policies. Structural equation modeling (SEM) was applied to survey data collected from 398 managers and experts in Chinese enterprises using established measures of AI functions, green dynamic capability, low-carbon policies, and low-carbon performance. The results show that green dynamic capability fully mediates the associations of predictive and decision-making AI with low-carbon performance. For automated AI, the indirect effect through green dynamic capability is significant, while the direct effect is at the borderline of statistical significance, suggesting a possible partial mediation pattern that should be interpreted cautiously. Low-carbon policies positively moderate the relationships between all three AI functions and low-carbon performance. These findings suggest that AI is associated with supply chain decarbonization partly through firms' green dynamic capabilities, while supportive policy conditions strengthen these relationships. The study therefore highlights the importance of integrating AI applications with organizational capabilities and supportive policy environments. Low-carbon performance in this study is perceived performance, reflecting respondents' subjective assessments rather than independently verified absolute emission figures.

Keywords: AI; Green dynamic capabilities; Low-carbon policies; Low-carbon performance; Low-carbon transition of supply chains

1. Introduction

Since global greenhouse gas emissions continue to rise, global ecosystems and economic development are increasingly vulnerable to the effects of global warming, carbon emissions, and carbon neutrality. Global warming, carbon emissions and carbon neutrality are pressing issues (Wang et al., 2023). According to the Special Report on Global Warming of 1.5 °C published by the Intergovernmental Panel on Climate Change (IPCC), the global temperature rise by 1.5 °C would decrease the risk of global warming for humans (IPCC, 2018). The results of the IPCC regarding 1.5 °C temperature control target emphasize the urgent opportunity for climate action. Despite pressure on global climate governance and decarbonization of supply chains, traditional supply chains, which depend on energy intensive models, suffer from low efficiency, data silos and delayed decision making. New technologies can be used to achieve “carbon reduction without losing efficiency” or even “carbon reduction while improving efficiency” (Choi et al., 2023).

Artificial intelligence (AI) has grown rapidly in recent years and it has also played a role in connection with reducing carbon emissions. AI can predict future carbon emissions based on current carbon reduction tasks and needs, providing guidance for enterprises to develop carbon reduction plans. Carbon emissions data collection AI

can automatically collect carbon footprint data using smart sensors on products and analyze the data (Miller et al., 2024). Packaging design AI and wireless sensor networks can improve energy saving and environmentally friendly packaging design, reduce the energy consumption rate of product packaging (Dai, 2023). Transportation AI is driving electric and autonomous vehicles, shared mobility and intelligent transportation systems, enabling improvements in the energy efficiency of transportation systems and contributing to reductions in greenhouse gas emissions (Miller et al., 2024). Overall, AI technology is targeted at different types of enterprises and its application in carbon reduction varies. However, overall, it concentrates on three dimensions: precise prediction based on data, optimal decision-making and control of automated equipment.

Since AI is focused on data analysis and decision support rather than physical carbon reduction technologies, its role in reducing carbon emissions is realized indirectly by empowering other industries to optimize their processes. Thus, for each node of the supply chain, building green dynamic capabilities through AI empowerment is the key step towards low carbon performance (Gan et al., 2024). Traditional supply chains tend to be linear in nature, with profit maximization above environmental factors. However, corporations, such as consumers, are demanding to companies not only focus on economic value but also on social value. Companies need to quickly identify key green resources to adapt to changes of external environment such as technological innovation and learn to integrate green dynamic capabilities to promote low carbon transformation of the supply chain to gain a competitive advantage. AI can contribute to enhancing green dynamic capabilities to pursue “carbon reduction without compromising efficiency” or even “carbon reduction while increasing efficiency”.

In addition to technical factors, external low carbon policies also have a complex effect on the process and effect of the low carbon transformation of the supply chain. It is known that this effect may be heterogeneous across regions, ownership types, and industry characteristics (Wang & Li, 2024), but ultimately the effect of policies will need to be determined through perceptions and behavior of enterprises. There are significant differences in perception of pressure or incentives from the same policy environment among different enterprises. This subjective perception is likely to influence the motivation and determination of enterprises to embrace new technologies for emission reduction and consequently influence the effect of empowerment of technological empowerment. Therefore, when promoting the collaborative mechanism of technology–policy–market and considering the different policy design at the macro level, it is more important to consider how policies can be transformed into the micro perception and actions of enterprises. Prior evidence shows that a firm’s low-carbon awareness promotes low-carbon behaviors, and this relationship is positively moderated by environmental legitimacy (Zhou et al., 2020). We will approach from the micro perspective of enterprises and empirically examine how the perceived policy pressure/incentive intensity moderates the process of AI technology empowering low carbon transformation. However, most existing studies focus on independent effects of AI or low carbon policies, and the micro-mechanisms by which both drive transformation in synergy are unclear. In particular, we have not found systematic theoretical explanations or test how AI influences the internal green management and innovation capabilities of enterprises, i.e. green dynamic capabilities. Second, theoretical evidences on whether external low carbon policies are direct driving factors or boundary conditions for enabling technologies are unclear. In this work, we constructed and tested mediated moderating model, to show how AI technology application is associated with the influence path of AI technology application green dynamic capability low carbon performance and define the moderating role of low carbon policies in this enabling process. This paper aims to deepen theoretical understanding of the collaborative mechanism of digital transformation and green transformation and to give precise empowerment and governance ideas for enterprises and policy makers in practice. In this study, the dependent variable is perceived low-carbon performance—respondents’ subjective assessments of their firms’ improvements in emission reduction and energy efficiency, rather than independently verified absolute emission figures.

2. Literature Review and Hypotheses

2.1 Artificial Intelligence and Low-Carbon Performance

The term AI was first proposed in 1956 at Dartmouth College (US) by McCarthy, Minsky, Newell, and Shannon. They aimed to learn how machines could mimic human intelligence. They defined AI as “science and engineering of creating intelligent machines”. Over the next 50 years, AI has received widespread attention and been applied across a variety of social domains. Haugeland (1989) defined AI as a machine capable of mimicking human thought and behavior, LeCun et al. (2015) declared AI should have independent rational thinking and action and Legg & Hutter (2007) emphasized that AI, as agent, can achieve goals in a dynamic environment, Makridakis (2017) saw AI as an automated tool capable of replacing human labor in routine tasks. Taddy (2018) included machine learning, logic processing and natural language processing, and Acemoglu & Restrepo (2018) saw AI as a tool that aids human work, capable to make autonomous decisions and monitor production equipment operations to improve efficiency, Kromann et al. (2020) highlighted that AI technological innovation plays a key role in driving rapid socioeconomic development.

In short, AI is a scientific discipline which uses technologies like deep learning and machine learning combined

with efficient programming and algorithm processing to simulate human cognitive functions like perception and decision making, assisting or replacing humans in automated tasks. Three key applications of AI are predictive capability, decision ability and automation. These three applications are also the three main aspects studied in this paper concerning AI impact on low carbon performance. The fundamental difference between AI and traditional digital technologies (e.g. ERP/MES information systems) is that traditional systems automate predefined processes and record data, while AI is capable of data-driven learning, reasoning and autonomous optimization: predictive AI detects hidden patterns from large amounts of unstructured data and makes forward-looking decisions; decision-making AI autonomously balances and optimizes choices in multi-objective scenarios; automated AI adjusts control strategies in dynamic environments.

Low carbon was formally introduced by the UK government in 2003 (UK Government, 2003). As global climate problems caused by carbon emissions became more severe, British government started calling for low carbon economy in 2003, which was popularized globally. Ang (1999) stated that changes in energy intensity are crucial for climate change; Pearson & Foxon (2012) examined low-carbon transition pathways using First Industrial Revolution, Kuo et al. (2015) suggested that companies should establish relevant systems and measures to reduce energy consumption, pollution, and carbon emissions, companies should adopt sustainable development practices (Zhang & Liu, 2020) and low-carbon performance evaluations reflect how well they have transitioned (Ali et al., 2021).

In short, low-carbon performance is the result of technological, managerial and institutional innovations of companies to reduce greenhouse gas emissions and energy efficiency in production, operations or services and also improve environmental, economic and social benefits. Low-carbon performance is divided into environmental, social and economic benefits and is aligned with the fundamental ESG. Recent research shows that AI for green and low-carbon development in supply chains is a new model and direction, focused on three areas: forecasting, decision-making and automation:

Predictive AI forecasts future trends by analyzing historical data using machine learning and deep learning. For example, it can predict carbon emissions (Wang & Li, 2024) and monitor changes using long short-term memory (LSTM) and statistical process control (Ezenkwu et al., 2024). Utilizing AI algorithms to predict energy consumption quantities (Shaik et al., 2024). Decision Making AI refers to intelligent systems that simulate or assist human decision making in complex environments using data modeling, optimization algorithms and multi-objective analysis (Wang & Li, 2024). For instance, AI algorithms can evaluate data from multiple sources to reduce routes, fuel consumption, and supply chain efficiency (Shaik et al., 2024). By integrating Factor Analysis (FA), Data Envelopment Analysis (DEA) and Analytic Hierarchy Process (AHP) (FA-DEA-AHP), this decision-making framework can improve the selection and selection of low-carbon suppliers and is theoretically useful for exploring coordinated development in low-carbon supply chains (He & Zhang, 2018).

Automated AI: Intelligent systems that use machine learning, rule engines, or robotic process automation (RPA) to automate repetitive tasks or complex business processes without human intervention. For example, AI can use smart sensors installed on products to automatically collect carbon footprint data in real time and analyze and process that data (Wang & Li, 2024). In order to improve market-oriented and eco-friendly design and eliminate pollutants in packaging design (Dai, 2023), AI is used to improve the environmental protection (IPCC, 2018; Yu et al., 2023). Automated Intelligent Robotics applications reduce energy consumption and labor requirements, increase output and reduce carbon emissions (Yu et al., 2023).

In summary, this paper proposes the following hypotheses:

H1: The application of AI technology is positively associated with low-carbon performance.

H1a: Predictive AI technologies are positively associated with low-carbon performance.

H1b: Decision-making AI technologies are positively associated with low-carbon performance.

H1c: Automated AI technologies are positively associated with low-carbon performance.

2.2 Green Dynamic Capabilities

Green dynamic capability is derived from dynamic capability theory proposed by Teece et al. (1997). Dynamic capability is the ability to rapidly adapt to changes in external environment; enterprises incorporating and reorganizing new internal and external resources can gain competitive advantage or maintain competitive advantage, thus rapid transformation. Technological innovations and environmental concerns have extended the concept to include technological innovation and environmental protection, hence concepts like digital dynamic capability or green dynamic capability. Chen & Chang (2013) first proposed the concept of green dynamic capability as an enterprise integrating and reorganizing existing resources to improve organizational environmental performance based on dynamic changes in environment, Huang et al. (2015) defined it from process perspective as an enterprise's ability to meet environmental requirements from government, society, consumers and markets by integrating internal and external resources in product design, production, management or service processes. This definition highlights resource integration and restructuring as an essential component of green dynamic capability. Compared to general dynamic capability theory, green dynamic capability emphasizes more on

recognizing, integrating and restructuring environment-related resources (knowledge, human capital, and technology) for both environmental benefits and economic benefits in dynamic situations. This paper is consistent with the theory proposed by Huang et al. (2015) defining green dynamic capability as an enterprise's capacity to develop green products in a sustainable development strategy.

Regarding the dimensions of this capability, the general view of dynamic capability research mainly follows the two paradigms of dynamic capability research: the holistic behavior view and the specific process view. Li (2014) constructed a theoretical model with four dimensions: green transformation, organizational flexibility, environmental insight, and learning ability. Mousavi et al. (2018) divided the green dynamic capabilities into three dimensions: environmental perception, resource acquisition and reconstruction according to dynamical nature of environment, Nkrumah et al. (2021) divided green dynamic capabilities according to the operation process of green supply chain: green supplier development, green manufacturing, marketing, and environmental participation. This paper agreed with Mousavi et al. (2018) and classifies the types of green dynamic capabilities according to perception ability, acquisition ability and reconstruction ability. According to dynamic capability theory, an enterprise's competitive advantage and outstanding performance ultimately result from its ability to integrate, construct and reconstruct internal and external resources (Teece et al., 1997). Applying this logic to the context of environmental management, green dynamic capabilities enable enterprises to continuously perceive green trends, acquire low-carbon technologies and resources, and restructure production and operation process. This constant adaptation and reshaping allow enterprises to adapt and restructure production and operation processes. This process allows enterprises to adapt and reshaping directly empowers them to adopt emission reduction technologies more effectively, improve energy structures, and develop low-carbon products and services (Huang et al., 2015). Existing research supports that green dynamic capabilities have a significant driving effect on enterprise environmental performance and green innovation (Huang et al., 2015). Green dynamic capabilities are therefore a key link between green strategic goals and green innovation (Huang et al., 2015).

Based on the above analysis, the following hypothesis is proposed:

H2: Green dynamic capabilities are positively associated with supply chain low-carbon performance.

AI is a digital empowerment tool for organizations. Its value is not only directly optimizing processes but also increasing efficiency and depth with which enterprises build and leverage green dynamic capabilities. Unlike traditional digital technologies such as ERP or MES, AI learns from data, draws inferences and creates optimization strategies. Predictive AI helps corporations to be sensitive to carbon constraints, green technologies and consumers preferences by analyzing large amounts of environmental data, policies and market information (Wang & Li, 2024). Decision-making AI uses multi-objective optimization algorithms and simulations to help enterprises to make informed decisions in complex cost and efficiency and carbon emissions trade-offs (Shaik et al., 2024). Automation AI uses IoT sensors and intelligent control systems to automate decision making processes, fundamentally reorganizing material and energy flows to institutionalize and scale green operations (Yu et al., 2023). Hence, AI technology empowers three fundamental dimensions of green dynamic capabilities: perception, acquisition and transformation.

Based on the above analysis, the following hypothesis is proposed:

H3: The application of AI technology is positively associated with green dynamic capabilities.

H3a: Predictive AI technologies are positively associated with green dynamic capabilities.

H3b: Decision-making AI technologies are positively associated with green dynamic capabilities.

H3c: Automated AI technologies are positively associated with green dynamic capabilities.

Based on the above analysis, AI technology is enabling green dynamic capability (H3), and green dynamic capability is the internal driving force of low-carbon performance (H2). Thus, green dynamic capability will play a central role in transmission. This theoretical deduction blends technology empowerment of AI, capacity building of enterprises and end-product performance into a framework that can be used to discover the micro-mechanism black box of digital transformation driving green transformation. It is important to point out that the complete mediation mechanism proposed here has important theoretical implications: AI technology itself does not directly reduce emissions, but must be transformed into substantive low-carbon performance by restructuring the organizational capabilities of enterprises, which helps them perceive green opportunities more sensitively, more effectively obtain green resources, and more flexibly redesign operations. This shifts AI from direct carbon reduction tool to capacity-building catalyst and provides insights into the relationship between technology and organizational change.

Theoretically, the existence of a fully mediating mechanism is inevitable. AI technology is information processing and decision support tool and its output is "information" and "optimization plans" rather than direct physical carbon reduction actions. Thus, AI value is achieved depends on transforming capabilities at the organizational level. The "information" level is the capacity foundation of the enterprise—AI must have enough green dynamic capabilities to transform carbon footprint predictions, logistics optimization plans or equipment control instructions into real emission reduction actions, otherwise AI output is "information" and cannot be translated into "actions", which is precisely the theoretical root of complete mediation mechanism. Finally, mediating effect is not unconditional and intensity is strongly determined by three boundary conditions: First, the

absorption capacity of the enterprise is the main limitation or reinforcement of the boundary conditions: enterprises with strong digital foundation and data literacy can translate AI outputs more efficiently to organizational learning and capability updates, and thus mediating effect will be more effective. Secondly, the management has to be aware of the environmental awareness and strategic commitment of the environment and strategic direction of the supply chain. If the management views AI as an efficiency tool with green strategic direction, the mediating effect will be more unimpeded. Thirdly, the technical characteristics of the industry and supply chain structure are important macro context conditions. Production process of process-oriented manufacturing industries (chemical-metallurgical industries) has strong continuity and complete energy consumption data, AI perception and automation functions can be easily converted into green process optimization. However, emission reduction scenarios in discrete manufacturing industries (e.g. electronic assembly and mechanical processing) are more scattered. The path to AI to build green capabilities may be more complex, and the intensity of the mediating effect may be more heterogeneous.

Based on the above analysis, the following hypothesis is proposed:

H4: Green dynamic capacity mediates the relationship between AI technology applications and low-carbon performance.

H4a: Green dynamic capacity mediates the relationship between predictive AI technology and low-carbon performance.

H4b: Green dynamic capacity mediates the relationship between decision-making AI technologies and low-carbon performance.

H4c: Green dynamic capacity mediates the relationship between automated AI technology and low-carbon performance.

2.3 Low-Carbon Policies

Low-carbon policy is used in this paper as an umbrella term for the regulatory, market-based, and fiscal instruments through which governments promote greenhouse gas reduction and low-carbon transition—principally emissions trading (Gan et al., 2024), carbon taxation (Lu et al., 2010), and low-carbon subsidies (Jiang et al., 2022). It includes the following three common tools: carbon trading policies, carbon tax policies, low carbon subsidy policies. Carbon pricing policies, tax incentives, or targets form the external environment and the core incentive-control mechanisms driving low carbon transitions of enterprises.

Carbon emissions trading policy (carbon emissions trading policy) establishes total emissions caps and transforms carbon emission rights into trading commodities by allowances. Market mechanisms guide corporate decisions on carbon reduction, promote the flow of market resources towards low carbon industries and drive low carbon transformation of supply chains. Carbon emissions trading system improves efficiency of corporate capital allocation, especially affecting firms with lower pollution levels and regions with stricter regulations. Carbon emissions trading system improves capital allocation efficiency by improving firm working capital management and asset utilization efficiency. Market activity and government efficiency play a positive moderator in carbon emissions trading system's effect on corporate capital allocation efficiency, with market activity and government efficiency are more effective combined promotional effect (Wang et al., 2022).

A carbon tax policy is a tax on carbon emissions. Under a carbon tax policy, a government sets different tax rates, taking a metric ton carbon dioxide equivalent as tax base, and turns carbon emissions into cost to businesses. Businesses can adjust production and operations, allocate resources and avoid high carbon taxes by reducing carbon emissions. Research has shown that carbon tax policies have a small effect on GDP, but yield significant carbon emission reduction results (Lu et al., 2010).

As part of the environmental economic policy mix, the low carbon subsidies policy uses money directly obtained from the tax incentives to reduce the cost for enterprises to innovate in low carbon technologies, focusing on two areas: clean energy infrastructure deployment and low carbon research and development. This policy reduces the barriers to companies developing or adopting new technologies, and also improves the competitiveness of low carbon technologies, and encourages enterprises to develop emissions reduction plans. Carbon emission reduction cost subsidies for suppliers, as well as green consumption subsidies promote low carbon production among manufacturers and improve the performance of the supply chain. When both kinds of subsidies are implemented, supply chain participants receive higher profits and products emit lower carbon emissions. a government policy which uses only carbon emission reduction cost subsidies for manufacturers is more effective in supporting a low carbon economy than one that uses only green consumption subsidies (Jiang et al., 2022).

Carbon trading, carbon taxes, and subsidies are the external institutional driving mechanism for corporate low carbon transformation. We treat these three instruments as a single low-carbon policy construct, following institutional theory's premise that organizations respond to the perceived institutional environment rather than to each regulatory instrument in isolation (DiMaggio & Powell, 1983). In this perspective, even if different policy tools work through different mechanisms—carbon taxes work through cost pressures, carbon trading through market incentives, subsidies by lowering innovation barriers—we therefore operationalize low-carbon policy as

the aggregate intensity of perceived policy pressure and incentives, rather than as instrument-specific measures. Prior evidence that policy effects on firm behavior are conditioned by firms' regulatory context supports this aggregated treatment (Jiang et al., 2022; Wang et al., 2022). Hence low-carbon policy operates as enterprises' subjective perception of global pressures and incentive strength, rather than as an objective statistical aggregate of individual policy tools (the institutional environment organizational perception strategic response (Wang et al., 2022)), but focus on micro-level mechanisms through which these policies influence corporate perceptions. In both Institutional Theory and Behavioral Decision Theory, these policies create differentiated pressures (e.g. carbon emission costs) and incentives (e.g. innovation subsidies) for enterprises (Jiang et al., 2022; Wang et al., 2022). The perceived pressure or incentive influences marginal benefits and motivations for enterprises to adopt AI technologies for emission reduction. Prior evidence shows that external information environments such as media attention moderate the effect of carbon risk on a firm's cost of debt financing, indicating that external institutional pressure influences firm behavior outcomes through shaping corporate perception (Zhou et al., 2018). High perceived pressures motivate companies to integrate AI into their operations and decision-making process to ensure emission reduction effectiveness, reduce costs, or generate benefits. Low perceived intensity may lead to lack of motivation for AI adoption. The perception intensity of low-carbon policy pressure and incentives positively modifies the relationship between AI adoption and low-carbon performance by strengthening the link between technology application and performance outcomes rather than directly impact emissions reduction efforts.

Based on the above analysis, the following hypothesis is proposed:

H5: Low-carbon policies positively moderate the association between AI technology adoption and low-carbon performance.

H5a: Low-carbon policies positively moderate the association between predictive AI technology and low-carbon performance.

H5b: Low-carbon policies positively moderate the association between decision-making AI technologies and low-carbon performance.

H5c: Low-carbon policies positively moderate the association between automated AI technologies and low-carbon performance.

The main objective of this paper is to advance existing theories in three ways: First, by describing functional aspects of AI technology and linking them with the three dimensions of green dynamic capabilities, it opens up a key micro-level mechanism through which digital transformation can lead to green transformation that has not been explored, filling a gap in research about relevant mediating processes; Second, it emphasizes the mediating role of green dynamic capabilities, highlighting dynamic importance of organizational capacity building in relation to technology itself for sustainable transformation; Third, by defining low-carbon policies as boundary conditions (moderating variables) which influence the "technology–capability–performance" link instead of direct predecessors, we provide a new theoretical view of complex interactions between external institutions and internal empowerment processes.

3. Research Model

A review of the relevant literature identifies several potential relationships among the application of AI technologies—predictive, decision-making, and automated AI—green dynamic capabilities, low-carbon performance, and low-carbon policies. Figure 1 presents a graphical representation of the research model and all associated hypotheses.

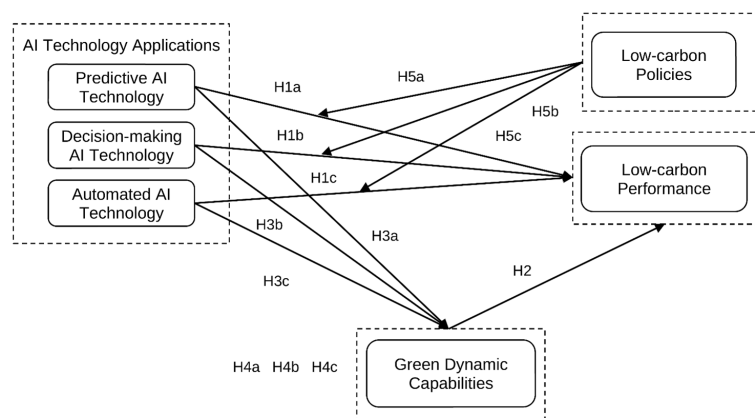


Figure 1. Research model

Among them, predictive AI, decision-making AI, and automated AI are independent variables, green dynamic capabilities serve as the mediating variable, low-carbon performance is the dependent variable, and low-carbon policies is the moderating variable.

4. Research Design

4.1 Questionnaire Design

The questionnaire was designed according to the hypotheses and using the model. The questionnaire is divided into four sections. The first section includes corporate characteristics and basic information (industry category, size, location, nature of company, respondent position). The second section measures the use of AI technology for low carbon transition: nine items based on three variables: predictive AI technology, decision AI technology and automation AI technology. Part Three measures the mediating variable (green dynamic capacity) and the moderating variable (low-carbon policy): 7 items. The three measures of low carbon policies: (1) “Our firm perceives strong pressure from carbon emission costs (e.g. carbon trading, carbon tax), (2) “Our firm perceives strong incentives from government low carbon innovation subsidies and similar measures”, (3) “Overall, our firm’s low carbon policy environment provides clear pressure or incentives for emission reduction.” The first two measures include both constraining (carbon costs) and incentivizing (subsidies) policies, while the third measures perceived policy intensity. Together, these three measures represent “overall perceived low carbon policy pressure/incentive intensity” as defined in this paper. Part four measures low-carbon performance; Part Two uses a five-point Likert scale.

To ensure the validity of the measurements, all constructs were measured according to standard scale revision and contextualization adaptation procedures. The steps for scale revision are: firstly, select mature scales that have been widely validated in international literature for each construct (score literature sources include: predictive AI and decision-making AI based Shaik et al. (2024), automated AI based on Yu et al. (2023), green dynamic capability based on Mousavi et al. (2018), low carbon performance based on Zhang & Liu (2020), low carbon policies based on Wang et al. (2022)). Two professors of supply chain management and three senior executives from enterprises are invited to form a group to review and revise specific expressions of items according to specific management practices and discourse system of China’s low carbon transformation of supply chains (Shaik et al., 2024). Third step is to collect feedback from small-scale pre-tests, fine-tuning to clarity and understandability of items, and form a contextualized and adaptable questionnaire for this paper.

4.2 Data Collection

The data are collected online and offline. The respondents were asked to evaluate actual AI systems deployed in their enterprises and in operation in their enterprises rather than generalized decisions on planned applications or general technical knowledge. Before each item of the questionnaire, we set guiding words which require respondents to answer according to actual AI technology and carbon management practices used in their enterprises. To ensure that respondents can identify and evaluate actual AI capabilities rather than general digital technology cognition, the questionnaire first provides definitions and specific examples of enterprise applications for the three types of functions: predictive AI, decision AI and automated AI in the introduction. In addition, respondents are required to confirm which types of AI applications their enterprises have deployed before answering and evaluate them according to their actual use experiences. Finally, the questionnaire design takes into account the perspective differences of respondents in management and technical positions, inviting both types of personnel to participate together. Management position focused on evaluating impact of AI on organization processes and resources and technical position focused on actual functions and performance of AI systems. For the external expert and other category of “External Expert” and “Other”, we have supply chain experts, industry analysts, and academic researchers who have long-term working relationships with the firms. They either know the firms AI systems and carbon management practices or have regular access to the firm data and meet the criteria. All data were cross-validated from different angles. 443 people participated, and 1 respondent per firm. We ensured the data was clear and normalized, and excluded invalid responses: 18 were excluded because we had more than 10% missing values on key variables, 16 for identical answers to all items (careless answering), 11 for standard residuals with absolute values greater than 3. 45 responses were excluded, with 398 valid responses and 89.84% recovery rate. The survey received informed consent from all participating firms and was exempt from ethics review as it involved only anonymous questionnaires without collecting personally identifiable information.

4.3 Descriptive Statistics

Table 1 shows that more than 50% of the samples are categorized as “Retail and Logistics” and 39.70% of samples sizes are less than 50 million. Samples from East China are 31.41% and Private Enterprises are 58.04%, while 12.31% are foreign funded samples. In terms of respondent information, more than 40% of samples are from

people in “Technical Positions”.

Table 1. Descriptive statistics

Name	Option	Frequency	Percentage (%)	Cumulative Percentage (%)
Industry category	Manufacturing	103	25.88	25.88
	Energy& Chemical	51	12.81	38.69
	Retail & Logistics	216	54.27	92.96
	Other	28	7.04	100
Enterprise size	<50 million	158	39.70	39.70
	50–500 million	146	36.68	76.38
	500 million–5 billion	56	14.07	90.45
	>5 billion	38	9.55	100
	Northeast China	53	13.32	13.32
Region	North China	79	19.85	33.17
	Central China	81	20.35	53.52
	East China	125	31.41	84.92
	Northwest China	13	3.26	88.19
	Southwest China	10	2.51	90.70
	South China	37	9.30	100
Enterprise type	State-owned enterprise	118	29.65	29.65
	Private enterprise	231	58.04	87.69
	Foreign-funded enterprise	49	12.31	100
	Management position	129	32.41	32.41
Respondent info	Technical position	164	41.21	73.62
	External expert	63	15.83	89.45
	Other	42	10.55	100
Total		398	100	100

5. Empirical Analysis

5.1 Reliability Analysis

This study assessed the reliability of the measurement scales using Cronbach’s α and corrected item-total correlation (CITC). As shown in Table 2, all Cronbach’s α coefficients exceed 0.8, indicating high internal consistency. All CITC values are above 0.4, confirming that each item is well correlated with its respective construct.

Confirmatory factor analysis (CFA) was conducted to evaluate the measurement model. The model fit indices are satisfactory: adjusted goodness-of-fit index (AGFI) = 0.945, normed fit index (NFI) = 0.958, incremental fit index (IFI) = 0.994, Tucker–Lewis index (TLI) = 0.992, comparative fit index (CFI) = 0.994, root mean square error of approximation (RMSEA) = 0.020. All factor loadings are significant and exceed 0.7, as detailed in Table 3, indicating that the items adequately represent their latent constructs. Convergent validity is assessed by average variance extracted (AVE) and composite reliability (CR). As reported in Table 3, all AVE values exceed 0.5 and all CR values exceed 0.8, demonstrating strong convergent validity.

Discriminant validity is assessed by comparing the square root of the AVE for each construct with its correlations with other constructs. As shown in Table 4, the diagonal elements in bold represent the square roots of AVE, while the off-diagonal elements are inter-construct correlations. For all constructs, the square root of AVE exceeds its highest correlation with any other construct, which indicates good discriminant validity. While the correlation coefficient between green dynamic capability and low carbon performance (0.762) is fairly high, square root of the AVE for green dynamic capability (0.762) equals this correlation, and the square root of AVE for low-carbon performance (0.764) is slightly larger, which indicates that the two concepts are distinguishable empirically. From theoretical point of view, green dynamic capability measures a firm’s process-oriented capability to “whether or not” it can integrate and restructure green resources, whereas low carbon performance measures a firm’s outcome-oriented output of “whether or not” it has achieved emission reduction results. The two concepts are distinguishable conceptually by a “capability–performance” logic and can therefore be treated as independent constructs.

5.2 Exploratory Factor Analysis

The Kaiser–Meyer–Olkin (KMO) measure and Bartlett’s test of sphericity were conducted to determine whether the variables were suitable for FA. As shown in Table 5, the KMO value was 0.885, and Bartlett’s test of sphericity was significant ($\chi^2 = 3764.925$, $df = 171$, $p < 0.001$), indicating sufficient inter-item correlation and suitability for

FA.

Table 2. Reliability analysis

Variable	Item	Corrected Item-Total Correlation (CITC)	Cronbach's α if Item Deleted	Cronbach's α Coefficient
Predictive AI technology	Predictive1	0.662	0.77	0.823
	Predictive2	0.681	0.752	
	Predictive3	0.689	0.744	
Decision-making AI technology	Decision1	0.708	0.762	0.836
	Decision2	0.718	0.752	
	Decision3	0.666	0.802	
Automated AI technology	Automated1	0.677	0.796	0.837
	Automated2	0.699	0.775	
	Automated3	0.724	0.750	
Green dynamic capabilities	Green dynamic capabilities1	0.693	0.801	0.847
	Green dynamic capabilities2	0.71	0.794	
	Green dynamic capabilities3	0.682	0.806	
	Green dynamic capabilities4	0.649	0.82	
Low-carbon performance	Low-carbon performance1	0.664	0.728	0.807
	Low-carbon performance2	0.64	0.753	
	Low-carbon performance3	0.663	0.728	
	Low-carbon policies1	0.796	0.815	
Low-carbon policies	Low-carbon policies2	0.748	0.857	0.884
	Low-carbon policies3	0.777	0.832	

Table 3. Factor loadings and convergent validity

Construct	Item	Factor Loading	AVE	CR
Predictive AI	Predictive1	0.744	0.607	0.822
	Predictive2	0.773		
	Predictive3	0.819		
Decision-making AI	Decision1	0.82	0.631	0.837
	Decision2	0.803		
	Decision3	0.759		
Automated AI	Automated1	0.769	0.633	0.838
	Automated2	0.802		
	Automated3	0.816		
Green dynamic capabilities	Green dynamic capabilities1	0.792	0.581	0.886
	Green dynamic capabilities2	0.766		
	Green dynamic capabilities3	0.769		
	Green dynamic capabilities4	0.719		
Low-carbon performance	Low-carbon performance1	0.769	0.584	0.808
	Low-carbon performance2	0.762		
	Low-carbon performance3	0.762		
	Low-carbon policies1	0.878		
Low-carbon policies	Low-carbon policies2	0.811	0.718	0.884
	Low-carbon policies3	0.852		

Note: AVE = average variance extracted; CR = composite reliability.

Table 4. Discriminant validity and correlation

Factor	Predictive AI	Decision AI	Automated AI	Green Dynamic Capabilities	Low-Carbon Performance	Low-Carbon Policies
Predictive AI technology	0.779					
Decision-making AI technology	0.309***	0.794				
Automated AI technology	0.396***	0.293***	0.796			
Green dynamic capabilities	0.611***	0.612***	0.543***	0.762		
Low-carbon performance	0.549***	0.539***	0.531***	0.762***	0.764	
Low-carbon policies	0.209***	0.239***	0.264***	0.274***	0.181**	0.847

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 5. KMO and Bartlett's test results

KMO Measure of Sampling Adequacy	0.885
Bartlett's Test of Sphericity	
Approx. Chi-square (χ^2)	3764.925
df	171
Sig.	<0.001

Note: KMO = Kaiser–Meyer–Olkin; df = degrees of freedom.

As shown in Table 6, five factors with eigenvalues greater than 1 were extracted, accounting for 69.868% of the cumulative variance. Before rotation, the first factor explained 35.595% of the variance, which is below the commonly used 40% threshold, indicating that a single factor cannot account for the majority of the item variance and that the risk of common method bias is low.

Table 6. Total variance explained in exploratory factor analysis (FA)

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative (%)	Total	% of Variance	Cumulative (%)	Total	% of Variance	Cumulative (%)
1	6.763	35.595	35.595	6.763	35.595	35.595	3.858	20.307	20.307
2	2.216	11.662	47.257	2.216	11.662	47.257	2.474	13.019	33.326
3	1.769	9.312	56.569	1.769	9.312	56.569	2.353	12.383	45.710
4	1.497	7.880	64.448	1.497	7.880	64.448	2.297	12.088	57.798
5	1.030	5.420	69.868	1.030	5.420	69.868	2.293	12.070	69.868
6	0.889	4.680	74.548						
7	0.522	2.749	77.297						
8	0.494	2.601	79.898						
9	0.474	2.495	82.393						
10	0.444	2.338	84.731						
11	0.401	2.113	86.844						
12	0.385	2.026	88.869						
13	0.369	1.944	90.814						
14	0.351	1.850	92.663						
15	0.325	1.708	94.372						
16	0.309	1.624	95.996						
17	0.284	1.496	97.492						
18	0.248	1.308	98.800						
19	0.228	1.200	100						

Note: extraction method: Principal Component Analysis.

As shown in Table 7, the loadings of each item on the expected dimension are relatively concentrated: the three dimensions of AI technology, low-carbon policies, and the relevant dimensions composed of green dynamic capabilities and low-carbon performance all present a clear preliminary structure. Subsequent CFA further tests the theoretical measurement model and construct discriminant validity.

Table 7. Rotated component matrix

Construct	Component				
	1	2	3	4	5
Predictive1				0.816	
Predictive2				0.826	
Predictive3				0.780	
Decision1			0.816		
Decision2			0.865		
Decision3			0.762		
Automated1					0.801
Automated2					0.808
Automated3					0.856
Green dynamic capabilities1	0.639				
Green dynamic capabilities2	0.651				
Green dynamic capabilities3	0.660				
Green dynamic capabilities4	0.594				
Low-carbon performance1	0.766				
Low-carbon performance2	0.718				
Low-carbon performance3	0.753				
Low-carbon policies1		0.898			
Low-carbon policies2		0.871			
Low-carbon policies3		0.892			

Note: extraction method: Principal Component Analysis. Rotation method: Varimax with Kaiser Normalization. Rotation converged in 6 iterations.

5.3 Measurement Model, Common Method Bias, and Discriminant Validity Testing

CFA was used to test the measurement model. Regarding common method bias, the first factor in the exploratory FA explained only 35.595% of the variance; further comparisons among six-factor, single-factor, and five-factor merged models were conducted to assess common method bias and the discriminant validity between green dynamic capabilities and low-carbon performance.

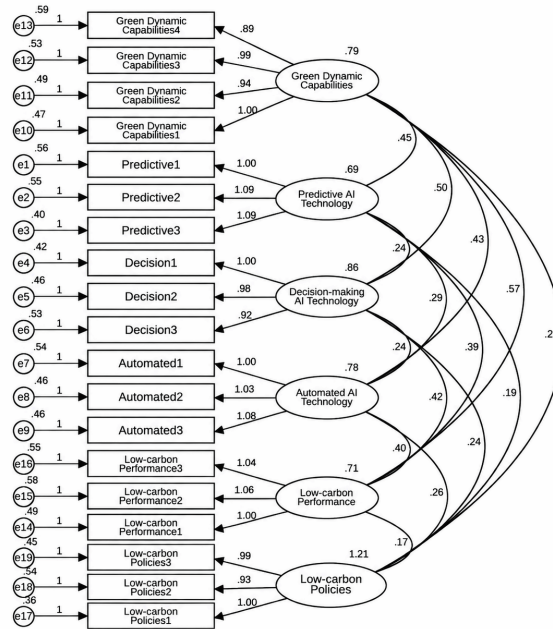


Figure 2. Six-factor confirmatory factor analysis (CFA) model

Figure 2 presents the specification of the six-factor measurement model. Predictive AI, decision-making AI, automated AI, green dynamic capabilities, low-carbon performance, and low-carbon policies are all estimated as mutually distinct latent variables.

Table 8 indicates that in the six-factor measurement model, the standardized loadings of all items range from 0.719 to 0.878, all exceeding the commonly used threshold, indicating good convergent measurement performance of the latent variables.

Table 8. Standardized factor loadings of the six-factor confirmatory factor analysis (CFA) measurement model

Path	Estimate	S.E.	C.R.	p	Std. Estimate
Predictive1 ← Predictive AI technology	1				0.744
Predictive2 ← Predictive AI technology	1.089	0.078	13.923	<0.001	0.773
Predictive3 ← Predictive AI technology	1.093	0.076	14.376	<0.001	0.819
Decision1 ← Decision-making AI technology	1				0.820
Decision2 ← Decision-making AI technology	0.979	0.062	15.851	<0.001	0.803
Decision3 ← Decision-making AI technology	0.916	0.06	15.137	<0.001	0.759
Automated1 ← Automated AI technology	1				0.769
Automated2 ← Automated AI technology	1.032	0.069	15.054	<0.001	0.802
Automated3 ← Automated AI technology	1.076	0.071	15.201	<0.001	0.816
Green dynamic capabilities1 ← Green dynamic capabilities	1				0.792
Green dynamic capabilities2 ← Green dynamic capabilities	0.944	0.06	15.756	<0.001	0.766
Green dynamic capabilities3 ← Green dynamic capabilities	0.988	0.062	15.84	<0.001	0.769
Green dynamic capabilities4 ← Green dynamic capabilities	0.894	0.061	14.655	<0.001	0.719
Low-carbon performance1 ← Low-carbon performance	1				0.769
Low-carbon performance2 ← Low-carbon performance	1.061	0.074	14.321	<0.001	0.762
Low-carbon performance3 ← Low-carbon performance	1.035	0.072	14.321	<0.001	0.762
Low-carbon policies1 ← Low-carbon policies	1				0.878
Low-carbon policies2 ← Low-carbon policies	0.928	0.049	18.904	<0.001	0.811
Low-carbon policies3 ← Low-carbon policies	0.987	0.05	19.863	<0.001	0.852

Note: S.E. = standard error; C.R. = critical ratio.

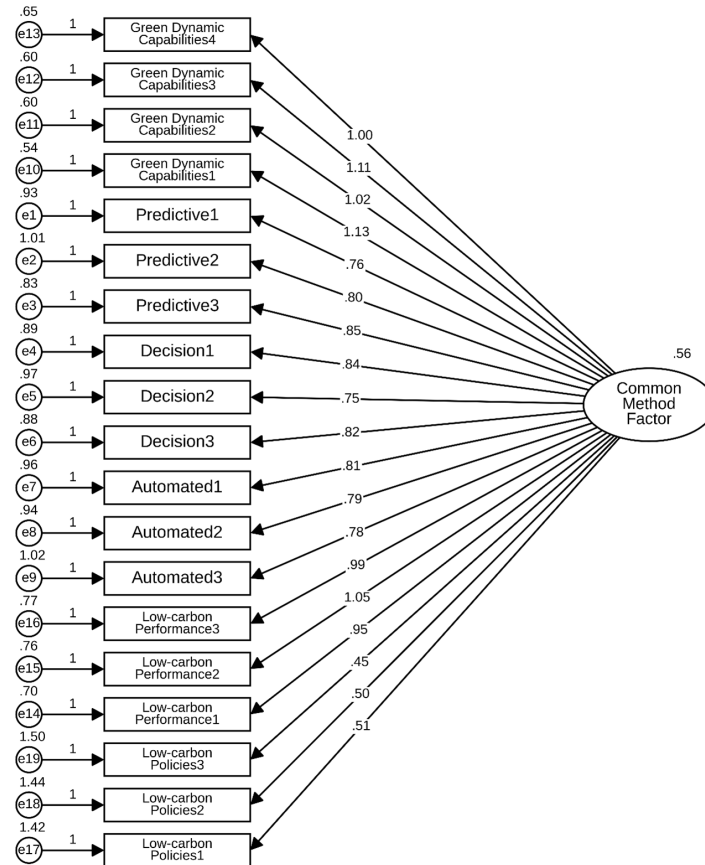
**Figure 3.** Single-factor confirmatory factor analysis (CFA) model

Figure 3 loads all observed items onto a single latent variable, used for comparison with the theoretical six-factor model and for testing common method bias.

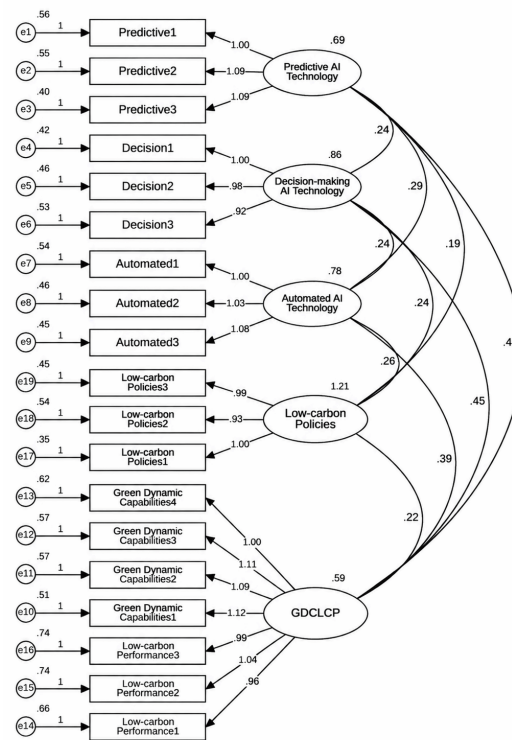
Table 9 presents the loading results when all items are constrained to a single latent variable. The fit of this competing model is compared with that of the six-factor model to assess common method bias.

Table 9. Standardized factor loadings of the single-factor confirmatory factor analysis (CFA) model

Path	Estimate	S.E.	C.R.	<i>p</i>	Std. Estimate
Green dynamic capabilities2 ← Single factor	1.024	0.08	12.769	<0.001	0.703
Green dynamic capabilities1 ← Single factor	1.126	0.083	13.601	<0.001	0.754
Predictive1 ← Single factor	0.755	0.08	9.422	<0.001	0.507
Predictive2 ← Single factor	0.803	0.084	9.551	<0.001	0.515
Predictive3 ← Single Factor	0.848	0.08	10.582	<0.001	0.574
Decision1 ← Single factor	0.84	0.082	10.282	<0.001	0.557
Decision2 ← Single factor	0.746	0.081	9.178	<0.001	0.494
Decision3 ← Single factor	0.815	0.081	10.094	<0.001	0.546
Automated1 ← Single factor	0.809	0.083	9.77	<0.001	0.527
Automated2 ← Single factor	0.794	0.082	9.706	<0.001	0.524
Automated3 ← Single factor	0.776	0.084	9.266	<0.001	0.499
Low-carbon performance3 ← Single factor	0.989	0.084	11.833	<0.001	0.647
Low-carbon performance2 ← Single factor	1.051	0.086	12.224	<0.001	0.67
Low-carbon performance1 ← Single factor	0.95	0.08	11.872	<0.001	0.649
Low-carbon policies3 ← Single factor	0.453	0.09	5.051	<0.001	0.267
Low-carbon policies2 ← Single factor	0.501	0.089	5.648	<0.001	0.299
Low-carbon policies1 ← Single factor	0.508	0.088	5.756	<0.001	0.305
Green dynamic capabilities4 ← Single factor	1				0.681
Green dynamic capabilities3 ← Single factor	1.115	0.084	13.276	<0.001	0.734

Note: The loading of the reference indicator was fixed at 1 for model identification. All estimated factor loadings were statistically significant at $p < 0.001$. S.E. = standard error; C.R. = critical ratio.

Figure 4 merges the items of green dynamic capabilities and low-carbon performance into a single latent variable to test the discriminant validity of the two constructs.

**Figure 4.** Five-factor merged confirmatory factor analysis (CFA) model

Note: GDCLCP = green dynamic capabilities and low-carbon performance.

Table 10 corresponds to the correct five-factor competing model: low-carbon policies remain independent, while the four items of green dynamic capabilities and the three items of low-carbon performance are jointly loaded onto the green dynamic capabilities and low-carbon performance factor. This model is used to test whether the two constructs can be merged.

Table 10. Standardized factor loadings of the five-factor merged confirmatory factor analysis (CFA) model

Path	Estimate	S.E.	C.R.	<i>p</i>	Std. Estimate
Predictive1 ← Predictive AI technology	1				0.744
Predictive2 ← Predictive AI technology	1.088	0.078	13.925	<0.001	0.773
Predictive3 ← Predictive AI technology	1.093	0.076	14.379	<0.001	0.819
Decision1 ← Decision-making AI technology	1				0.82
Decision2 ← Decision-making AI technology	0.979	0.062	15.849	<0.001	0.802
Decision3 ← Decision-making AI technology	0.916	0.061	15.14	<0.001	0.759
Automated1 ← Automated AI technology	1				0.768
Automated2 ← Automated AI technology	1.032	0.069	15.042	<0.001	0.802
Automated3 ← Automated AI technology	1.077	0.071	15.201	<0.001	0.817
Low-carbon policies1 ← Low-carbon policies	1				0.879
Low-carbon policies2 ← Low-carbon policies	0.927	0.049	18.898	<0.001	0.811
Low-carbon policies3 ← Low-carbon policies	0.985	0.05	19.851	<0.001	0.851
Green dynamic capabilities2 ← GDCLCP	1.032	0.078	13.309	<0.001	0.726
Green dynamic capabilities1 ← GDCLCP	1.123	0.08	14.069	<0.001	0.771
Low-carbon performance3 ← GDCLCP	0.985	0.081	12.178	<0.001	0.66
Low-carbon performance2 ← GDCLCP	1.04	0.083	12.523	<0.001	0.68
Low-carbon performance1 ← GDCLCP	0.96	0.078	12.384	<0.001	0.672
Green dynamic capabilities4 ← GDCLCP	1				0.698
Green dynamic capabilities3 ← GDCLCP	1.108	0.081	13.688	<0.001	0.748

Note: GDCLCP = green dynamic capabilities and low-carbon performance; S.E. = standard error; C.R. = critical ratio.

Table 11 shows that the six-factor model fits best ($\chi^2/\text{df} = 1.162$, CFI = 0.994, TLI = 0.992, RMSEA = 0.020), while the single-factor model fits very poorly (CFI = 0.578, RMSEA = 0.160). Combined with the finding that the first factor explains only 35.595% of the variance, common method bias is not a major threat. After merging green dynamic capabilities and low-carbon performance, the fit deteriorates significantly compared with the six-factor model ($\Delta\chi^2 = 108.743$, $\Delta\text{df} = 5$, $p < 0.001$), providing supplementary evidence for the discriminant validity of the two constructs.

Table 11. Comparison of fit indices of confirmatory factor analysis (CFA) competing models

Model	χ^2	df	χ^2/df	GFI	CFI	TLI	RMSEA
Six-factor model	159.193	137.000	1.162	0.960	0.994	0.992	0.020
Single-factor model	1696.698	152.000	11.162	0.673	0.578	0.526	0.160
Five-factor merged model	267.936	142.000	1.887	0.929	0.966	0.959	0.047

Note: df = degrees of freedom; GFI = goodness-of-fit index; CFI = comparative fit index; TLI = Tucker–Lewis index; RMSEA = root mean square error of approximation.

As shown in Table 12, the heterotrait–monotrait ratio (HTMT) values for all construct pairs are below 0.85, and the Bootstrap 95% confidence intervals do not include 1, indicating acceptable discriminant validity.

5.4 Structural Equation Modeling

This study employed structural equation modeling (SEM) using AMOS 26 to test the hypothesized relationships. The overall model fit is satisfactory: $\chi^2/\text{df} = 1.336$, root mean square residual (RMR) = 0.037, GFI = 0.963, AGFI = 0.946, NFI = 0.960, IFI = 0.990, TLI = 0.987, CFI = 0.989, RMSEA = 0.029. Figure 5 presents the structural equation model with path coefficients. Table 13 reports the path coefficients and their significance.

As shown in Table 13, all three AI functions are significantly positively associated with green dynamic capabilities, supporting H3a, H3b, and H3c. Green dynamic capabilities, in turn, are positively associated with low-carbon performance, supporting H2. Furthermore, the direct associations of predictive, decision-making, and automated AI with low-carbon performance are also significant, providing support for H1a, H1b, and H1c.

5.5 Mediation Effects

Mediation effects were tested using 5,000 Bootstrap resamples and bias-corrected (BC) 95% confidence intervals. Table 14 simultaneously reports the estimates and confidence intervals of total effects, direct effects, and indirect effects. The confidence intervals of the indirect effects of the three types of AI technology on low-carbon performance through green dynamic capabilities all exclude zero. The confidence intervals of the direct effects of predictive and decision-making AI include zero; the lower bound of the direct effect of automated AI is zero. Green dynamic capabilities play a significant mediating role in the associations between the three types of AI technology and low-carbon performance. The direct effects of predictive AI and decision-making AI are not

significant, which is consistent with full mediation. For automated AI, the indirect effect is significant, whereas the direct effect is at the borderline of statistical significance ($p = 0.050$, 95% confidence interval [0.000, 0.320]). Therefore, the results suggest a possible partial mediation pattern for automated AI, but this interpretation should be treated cautiously.

Table 12. HTMT discriminant validity test and bootstrap confidence intervals

Construct 1	Construct 2	HTMT	95% CI Lower	95% CI Upper	Conclusion
Predictive AI technology	Decision-making AI technology	0.312	0.169	0.458	Supported
Predictive AI technology	Automated AI technology	0.399	0.253	0.544	Supported
Predictive AI technology	Green dynamic capabilities	0.606	0.478	0.728	Supported
Predictive AI technology	Low-carbon performance	0.543	0.410	0.680	Supported
Predictive AI technology	Low-carbon policies	0.211	0.078	0.350	Supported
Decision-making AI technology	Automated AI technology	0.297	0.149	0.454	Supported
Decision-making AI technology	Green dynamic capabilities	0.614	0.485	0.742	Supported
Decision-making AI technology	Low-carbon performance	0.545	0.416	0.676	Supported
Decision-making AI technology	Low-carbon policies	0.243	0.113	0.375	Supported
Automated AI technology	Green dynamic capabilities	0.546	0.411	0.677	Supported
Automated AI technology	Low-carbon performance	0.536	0.397	0.670	Supported
Automated AI technology	Low-carbon policies	0.267	0.127	0.401	Supported
Green dynamic capabilities	Low-carbon performance	0.761	0.640	0.881	Supported
Green dynamic capabilities	Low-carbon policies	0.273	0.136	0.407	Supported
Low-carbon performance	Low-carbon policies	0.179	0.067	0.324	Supported

Note: HTMT = heterotrait–monotrait ratio; CI = confidence interval.

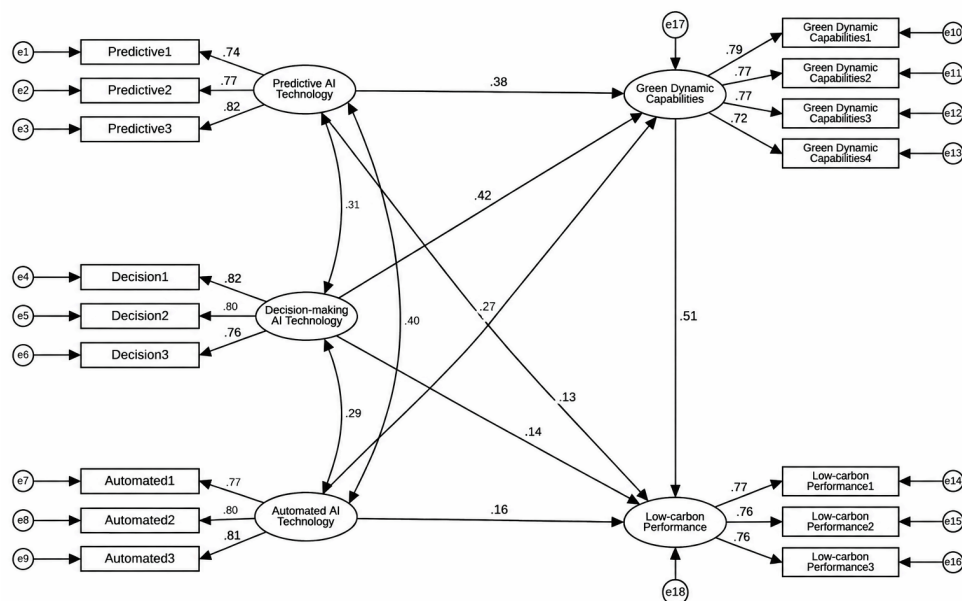


Figure 5. Structural equation model diagram (path and mediation effect analysis)

Table 13. Path analysis results

Path	Estimate	S.E.	C.R.	<i>p</i>	Std.	Hypothesis
Green dynamic capabilities ← Predictive AI technology	0.401	0.059	6.807	<0.001	0.375	Support H3a
Green dynamic capabilities ← Decision-making AI technology	0.398	0.05	8.001	<0.001	0.416	Support H3b
Green dynamic capabilities ← Automated AI technology	0.273	0.052	5.244	<0.001	0.273	Support H3c
Low-carbon performance ← Green dynamic capabilities	0.482	0.089	5.431	<0.001	0.508	Support H2
Low-carbon performance ← Predictive AI technology	0.133	0.065	2.045	0.041	0.131	Support H1a
Low-carbon performance ← Decision-making AI technology	0.127	0.058	2.19	0.029	0.140	Support H1b
Low-carbon performance ← Automated AI technology	0.154	0.055	2.787	0.005	0.162	Support H1c

Note: S.E. = standard error; C.R. = critical ratio.

Table 14. Bootstrap mediation effect test results

Mediation Path	Effect Type	Effect Estimate	95% CI Lower	95% CI Upper	<i>p</i>
Predictive AI technology → Low-carbon performance	Direct effect	0.133	-0.044	0.326	0.128
Predictive AI technology → Green dynamic capabilities → Low-carbon performance	Indirect effect	0.193	0.081	0.379	<0.001
Predictive AI technology → Low-carbon performance	Total effect	0.327	0.184	0.492	<0.001
Decision-making AI technology → Low-carbon performance	Direct effect	0.127	-0.037	0.299	0.117
Decision-making AI technology → Green dynamic capabilities → Low-carbon performance	Indirect effect	0.192	0.082	0.362	<0.001
Decision-making AI technology → Low-carbon performance	Total effect	0.319	0.195	0.465	<0.001
Automated AI technology → Low-carbon performance	Direct effect	0.154	0.000	0.320	0.050
Automated AI technology → Green dynamic capabilities → Low-carbon performance	Indirect effect	0.132	0.050	0.281	<0.001
Automated AI technology → Low-carbon performance	Total effect	0.286	0.160	0.430	<0.001

Note: CI = confidence interval.

5.6 Moderation Effects

The moderating variable low-carbon policies was operated as an observed variable (measured by its mean items) and hierarchical regression can better understand the incremental explanatory power (R^2) and simple slopes of observed moderating variables. Therefore, hierarchical regression was used to test the moderating effect of low-carbon policies. Model 1 included firm size; Model 2 added mean-centered AI technology variables and low-carbon policies variables; Model 3 further added their interaction term. Each table fully reports the coefficients, collinearity diagnostics, R^2 , ΔR^2 , F values, and ΔF for each step.

As shown in Table 15, after adding predictive AI, low-carbon policies, and their interaction term, the model's explanatory power increased from .202 to .233 ($\Delta R^2 = 0.032$, $\Delta F = 16.208$, $p < 0.001$). The interaction term was significantly positive ($\beta = 0.183$, $p < 0.001$), and all VIF values were below 1.10, supporting the positive moderating effect of low-carbon policies.

As shown in Table 16, after adding the interaction term between decision-making AI and low-carbon policies, the model R^2 increased from 0.204 to 0.227 ($\Delta R^2 = 0.023$, $\Delta F = 11.583$, $p < 0.001$). The interaction term was significantly positive ($\beta = 0.162$, $p < 0.001$), and no obvious multicollinearity problem existed, supporting the positive moderating effect of low-carbon policies.

Table 15. Moderating effect of low-carbon policies on the relationship between predictive AI technology and low-carbon performance

Variable	Unstandardized Coefficient		Standardized Coefficient			Collinearity Diagnostics	
	<i>B</i>	Std. Error	β	<i>t</i>	<i>p</i>	VIF	Tolerance
Model 1: Control variable							
Constant	3.4	0.138	–	24.721	<0.001	–	–
Firm size	0.003	0.047	0.003	0.057	0.954	1	1
R^2				0			
Adjusted R^2				-0.003			
ΔR^2				–			
ΔF				–			
F				$F(1, 396) = 0.003, p = 0.954$			
D-W statistic				1.889			
Model 2: Adding main effects							
Constant	3.41	0.124	–	27.606	<0.001	–	–
Firm size	-0.001	0.043	-0.001	-0.025	0.98	1.006	0.994
Predictive AI Technology (centered)	0.428	0.046	0.43	9.388	<0.001	1.034	0.967
Low-carbon policies (centered)	0.064	0.039	0.075	1.631	0.104	1.04	0.962
R^2				0.202			
Adjusted R^2				0.196			
ΔR^2				0.202			
ΔF				$\Delta F(2, 394) = 49.787, p < 0.001$			
F				$F(3, 394) = 33.193, p < 0.001$			
D-W statistic				1.888			
Model 3: Adding interaction term							
Constant	3.38	0.121	–	27.833	<0.001	–	–
Firm size	0	0.042	0	0.008	0.993	1.006	0.994
Predictive AI technology (centered)	0.469	0.046	0.471	10.225	<0.001	1.088	0.919
Low-carbon policies (centered)	0.062	0.038	0.073	1.618	0.106	1.04	0.961
Predictive AI technology $\times$ Low-carbon policies	0.132	0.033	0.183	4.026	<0.001	1.054	0.949
R^2				0.233			
Adjusted R^2				0.226			
ΔR^2				0.032			
ΔF				$\Delta F(1, 393) = 16.208, p < 0.001$			
F				$F(4, 393) = 29.908, p < 0.001$			
D-W statistic				1.956			

Note: VIF = variance inflation factor; A dash (–) indicates not applicable.

Table 16. Moderating effect of low-carbon policies on the relationship between decision-making AI technology and low-carbon performance

Variable	Unstandardized Coefficient		Standardized Coefficient			Collinearity Diagnostics	
	<i>B</i>	Std. Error	β	<i>t</i>	<i>p</i>	VIF	Tolerance
Model 1: Control variable							
Constant	3.4	0.138	–	24.721	<0.001	–	–
Firm size	0.003	0.047	0.003	0.057	0.954	1	1
<i>R</i> ²			0				
Adjusted <i>R</i> ²			-0.003				
ΔR^2			–				
ΔF			–				
<i>F</i>			<i>F</i> (1, 396) = 0.003, <i>p</i> = 0.954				
D-W statistic			1.889				
Model 2: Adding main effects							
Constant	3.458	0.123	–	28.027	<0.001	–	–
Firm size	-0.019	0.043	-0.02	-0.439	0.661	1.007	0.993
Decision-making AI Technology (centered)	0.43	0.045	0.435	9.457	<0.001	1.046	0.956
Low-carbon policies (centered)	0.054	0.039	0.063	1.36	0.174	1.051	0.952
<i>R</i> ²			0.204				
Adjusted <i>R</i> ²			0.198				
ΔR^2			0.204				
ΔF			ΔF (2, 394) = 50.461, <i>p</i> < 0.001				
<i>F</i>			<i>F</i> (3, 394) = 33.642, <i>p</i> < 0.001				
D-W statistic			1.986				
Model 3: Adding interaction term							
Constant	3.408	0.123	–	27.79	<0.001	–	–
Firm size	-0.01	0.042	-0.011	-0.246	0.806	1.01	0.99
Decision-making AI technology (centered)	0.489	0.048	0.494	10.167	<0.001	1.2	0.833
Low-carbon policies (centered)	0.045	0.039	0.053	1.163	0.245	1.055	0.948
Decision-making AI technology × Low-carbon policies	0.117	0.034	0.162	3.403	<0.001	1.152	0.868
<i>R</i> ²			0.227				
Adjusted <i>R</i> ²			0.219				
ΔR^2			0.023				
ΔF			ΔF (1, 393) = 11.583, <i>p</i> < 0.001				
<i>F</i>			<i>F</i> (4, 393) = 28.805, <i>p</i> < 0.001				
D-W statistic			1.968				

Note: VIF = variance inflation factor; A dash (–) indicates not applicable.

As shown in Table 17, after adding the interaction term between automated AI and low-carbon policies, the model *R*² increased from 0.198 to 0.221 ($\Delta R^2 = 0.024$, $\Delta F = 11.972$, *p* < 0.001). The interaction term was significantly positive ($\beta = 0.165$, *p* < 0.001), further supporting the positive moderating effect of low-carbon policies. Thus, H5a, H5b, and H5c are supported.

Table 17. Moderating effect of low-carbon policies on the relationship between automated AI technology and low-carbon performance

Variable	Unstandardized Coefficient		Standardized Coefficient			Collinearity Diagnostics	
	<i>B</i>	Std. Error	β	<i>t</i>	<i>p</i>	VIF	Tolerance
Model 1: Control variable							
Constant	3.4	0.138	—	24.721	<0.001	—	—
Firm size	0.003	0.047	0.003	0.057	0.954	1	1
R^2				0			
Adjusted R^2				-0.003			
ΔR^2				—			
ΔF				—			
F				$F(1, 396) = 0.003, p = 0.954$			
D-W statistic				1.889			
Model 2: Adding main effects							
Constant	3.441	0.124	—	27.785	<0.001	—	—
Firm size	-0.012	0.043	-0.013	-0.289	0.773	1.006	0.994
Automated AI technology (centered)	0.416	0.045	0.429	9.256	<0.001	1.056	0.947
Low-carbon policies (centered)	0.046	0.04	0.054	1.167	0.244	1.062	0.942
R^2				0.198			
Adjusted R^2				0.192			
ΔR^2				0.198			
ΔF				$\Delta F(2, 394) = 48.529, p < 0.001$			
F				$F(3, 394) = 32.354, p < 0.001$			
D-W statistic				1.853			
Model 3: Adding interaction term							
Constant	3.411	0.122	—	27.861	<0.001	—	—
Firm size	-0.013	0.042	-0.013	-0.3	0.764	1.006	0.994
Automated AI technology (centered)	0.476	0.048	0.491	9.999	<0.001	1.219	0.82
Low-carbon policies (centered)	0.036	0.039	0.042	0.91	0.363	1.068	0.936
Automated AI technology $\times$ Low-carbon policies	0.116	0.033	0.165	3.46	<0.001	1.155	0.866
R^2				0.221			
Adjusted R^2				0.213			
ΔR^2				0.024			
ΔF				$\Delta F(1, 393) = 11.972, p < 0.001$			
F				$F(4, 393) = 27.934, p < 0.001$			
D-W statistic				1.846			

Note: VIF = variance inflation factor; A dash (—) indicates not applicable.

6. Conclusions and Recommendations

6.1 Main Findings

Based on data from 398 valid responses (from 398 distinct firms), we found that the use of AI (predictive, decision-making, automation) is positively associated with the low-carbon performance of supply chains. More importantly, the study found that green dynamic capabilities play a mediating role in this association process. Specifically, green dynamic capabilities fully mediate the associations of predictive and decision-making AI with low-carbon performance, while partially mediating the association of automated AI. The positive associations of

predictive and decision-making AI are observed primarily through the enhancement of green dynamic capabilities, whereas automated AI exhibits both an indirect association via green dynamic capabilities and a small but significant direct association with operational efficiency. Moreover, low-carbon policy pressures and incentives that firms perceive positively moderate the association between AI technology use and low-carbon performance. It should be noted that low-carbon performance in this study refers to perceived low-carbon performance—respondents' subjective assessments of their firms' improvements in emission reduction and energy efficiency, rather than independently verified absolute emission figures.

6.2 Theoretical Implications

The findings of this study offer three insights for theoretical development in related fields: First, we have explored and expanded the theory of dynamic capabilities in relation to sustainable development and digital transformation. We have shown that green dynamic capabilities represent the main medium and theoretical “black box” for understanding digital and intelligent technologies interactions with environmental performance. We conclude that the core of a company's ability to link the potential of technologies such as AI to real low-carbon performance is associated with the development and nurturing of dynamic capabilities specific to green transformation. Unlike previous work that treated AI as a single technology or discussed its emission reduction effects broadly, our contributions are threefold: we disaggregate AI into three functions (predictive, decision-making, and automation) and demonstrate their respective associations with low-carbon performance; we consider low-carbon policies (rather than just direct predecessors) as a moderator of the “technology–capability–performance” chain; and we provide Chinese supply chain firms with evidence of the synergy between digital and green transformation in developing countries.

Second, we propose to understand how AI can be used to be associated with improved organizational performance. By classifying AI technologies in three functions (predictive, decision-making, automation) and their distinct effects, we overcome the analytical limitation of treating AI as a single technology and provide a more refined theoretical framework and measurement base for future work.

Finally, we combine technology empowerment and institutional theory. By examining the moderating role of low-carbon policies, we show that corporate low-carbon transition is a process of co-evolution and interaction between external pressure/incentives and internal technological capacity building, which offers new theoretical insights and empirical evidence on the micro-level mechanisms of “technology–institutional” synergy associated with sustainable transformation. From a micro-level perspective of sustainable transformation, we provide empirical evidence consistent with a sociotechnical transition perspective, in which external institutional pressure (low-carbon policies) amplifies the technology-enablement effects by shaping the development of internal firm capabilities. This interaction chain presents new analytical frameworks and empirical evidence on multi-level interactions in sustainable transformation. Climate mitigation, we highlight AI's unique role as a “capacity building catalyst” rather than a direct carbon reduction tool, and micro-level evidence of the indirect but notably associated link between digital technologies and greenhouse gas emission reduction: enterprises must first combine AI outputs with significant improvements in organizational capabilities before observing measurable emissions reduction outcomes. This finding informs the assessment of AI's potential value in climate action. In terms of environmental governance, we show how low-carbon policies are associated with technological empowerment through mediating role of corporate perception, and highlight a broad governance chain that spans policy signals corporate perception capability development environmental performance. This provides the theoretical basis for decentralized and diversified environmental governance models in the digital age: effective environmental governance relies not only on top-down policy regulation, but also on corporate perception and response mechanisms, and AI technology provides the data base and decision support needed to close-loop operation. It should be noted that the conclusions of this study are based on perceived performance data and reflect associations among the constructs rather than causal inferences; the dependent variable is perceived low-carbon performance, rather than independently verified actual emission reductions.

6.3 Practical Implications

Based on the “AI–Competency–Performance” association pathway, this study offers the following actionable recommendations for business managers:

Implement different AI investment and integration strategy. Companies should recognize that different AI technologies are associated with different capability dimensions. They should prioritize prescriptive AI (such as supply chain carbon optimization algorithms) to be able to take green decisions directly, while predictive AI (such as carbon footprint modeling) is associated with improved environmental insights and the development of dynamic green capabilities. Utilize AI projects as a tool for developing green organizational capacity. Companies should not view AI as a technical tool but rather as an organizational transformation initiative. AI should drive data integration, process reengineering, and collaboration between departments, transforming the process of integrating technology into a process that develops green learning, integration and restructuring capacity.

Properly use AI in supply chain collaboration. Core enterprises can use AI-powered sharing platforms to exchange carbon emissions data with suppliers and collaborate on low carbon technologies, and extend their green capabilities from the enterprise to supply chain network and improve the low carbon performance of the supply chain. It should be noted that the above recommendations are based on the associative evidence from this study, and managers should verify and apply them in their specific firm contexts.

6.4 Policy Implications

To more effectively leverage the “technology–policy” synergy, policymakers may consider:

Develop different policies for each stage of AI adoption. For industries or regions in the early stages of AI adoption, subsidies should be provided for AI infrastructure and dissemination of knowledge on low-carbon technologies; for enterprises already established, binding policies such as carbon markets and carbon taxes should be strengthened in order to encourage them to integrate their existing AI capabilities in emissions reduction efforts, increasing regulatory impact.

Develop guidelines for a supply chain transition. Lead enterprises should disclose their performance in reducing supply chain carbon emissions by AI and support AI-based green supply chain innovation projects, using technology of core enterprises to drive low carbon supply chain transition of upstream and downstream small and medium enterprises.

The use of AI tools to improve policy regulation and service delivery. Government regulators can explore AI algorithms for automated monitoring, reporting and verification (MRV) carbon emissions data and build AI-based public carbon management service platforms for intelligent carbon diagnostics and emissions reduction pathway simulations for businesses to reduce the cost of policy compliance and technological innovation.

6.5 Limitations and Future Research

Although this study has made some contributions, there are still several limitations that need to be addressed in future research.

First, this study relies solely on respondents’ subjective perceptions to measure low-carbon performance, rather than objective, third-party verified emissions data or corporate carbon accounting records. Although perceptual measures are widely used in organizational behavior and strategic management research, we have taken necessary steps to ensure reliability and validity, perceived improvements may not correspond fully to actual absolute emissions reductions. Therefore, the results of this study should be interpreted as evidence of significant positive correlation between AI adoption and perceived improvements in low-carbon performance, rather than as evidence of a reduction of absolute carbon emissions. Future work could include objective corporate-level measures such as carbon intensity per unit of output or Carbon Disclosure Project (CDP) scores. Self-reporting instruments may also be subject to common-method bias and social-desirability bias. Although we have taken procedural safeguards such as anonymity, reverse-scored items and Harman’s one-way test, future work could further strengthen causal inferences by using multiple sources such as matching AI usage data with independently audited emissions reports or using longitudinal tracking measures.

Second, we use a cross-sectional survey design where all variables are collected at one time point. We cannot draw strong causal conclusions. The models we propose are well theoretically based and the results on mediating effects are realistically expected, but causal testing still requires longitudinal panel data or quasi-experimental design. Third, “transition” implies a gradual, dynamic change of organizational practices, capabilities and strategies over time. However, our cross-sectional snapshot can only show contemporaneous associations between variables, and cannot reveal dynamic evolutions of capability accumulation, technological evolution and performance improvement. We strongly encourage future work to use multi-time-point data collection, longitudinal panel design or process-oriented case studies to better understand how AI enabled green transitions actually occur over time.

Third, since the sample is entirely Chinese, we ask if we can generalize our findings to other countries. China is a state-led country with distinct features such as a national carbon trading market, a tier-up process and implementation of the “dual carbon” goal by local governments, and state-owned enterprises. China has also played a key role in global supply chains and a unique digitalization path such as “Internet Plus” and “Made in China 2025” that may affect the relationship between AI and green capabilities in ways that are not universally applicable. We caution when generalize our results to regions with different institutional logics and recommend cross-national replication studies in various institutional contexts such as Europe, North America, and Southeast Asia to test the cross-context robustness of our results.

Fourth, we recognize limitations in the disclosure of measurement instruments. Due to confidentiality agreements with participating companies, we cannot disclose all survey items. Although we detailed scale selection and adaptation process and provided representative items—all scales have passed reliability and validity tests—this limitation does not guarantee reproducibility of the study. Future researchers may publish the complete measurement instruments (where allowed) or adopt industry standard publicly available measurement scales to

accumulate knowledge and compare studies.

Fifth, due to practical constraints, we were unable to examine the potential moderating effects of firm type, size, or ownership structure. Future research could employ a multi-group analysis approach to explore these heterogeneity effects in greater depth, thereby providing more targeted guidance for different types of firms. Furthermore, while we clarified the eligibility criteria for external experts and respondents categorized as “Other”, we did not implement a distinct sensitivity analysis to exclude these two subgroups. Such robustness checks could be undertaken in future studies. We also combine carbon trading, carbon taxes, and subsidies into a single “low-carbon policies” that captures the perceived pressure/incentive intensity of the firm. However, we acknowledge that these three policy instruments may operate by different economic and behavior mechanisms (cost constraints, innovation subsidies, market signals), and their moderating effects may differ in direction and strength. Future work should consider the moderating roles of each policy instrument separately to provide more specific guidance on policy design. We note that these three measurement items are not designed to analyze each policy tool independently; rather, the first two items capture cost constraint and incentive subsidy policy signals and the third item aggregates them into an overall perceived intensity. This design aligns with our theoretical focus—how firms combine scattered policy signals into a coherent perception of pressure/incentive that influences their decision to adopt AI.

By addressing the limitations mentioned above, future research can build on the findings of this study to provide more compelling evidence on how to effectively leverage AI to be associated with corporate low-carbon transitions.

In summary, we identify positive associations between the three AI functions (predictive, decision-making, automation) and supply chain low-carbon performance, mediated by green dynamic ability and moderated by low-carbon policies. Based on survey data from Chinese firms, this paper provides empirical reference for understanding the synergy between digital technologies and green transformation. The dependent variable is perceived low-carbon performance. All conclusions should be read as associative evidence derived from cross-sectional data, rather than causal inferences.

Author Contributions

Conceptualization, Y.K., Y.Y., and Y.C.; methodology, Y.Y. and Y.K.; software, Y.Y.; validation, Y.Y. and Y.K.; formal analysis, Y.Y.; investigation, Y.Y. and Y.C.; resources, Y.K.; data curation, Y.Y.; writing—original draft preparation, Y.C.; writing—review and editing, Y.C., Y.Y., and Y.K.; visualization, Y.Y.; supervision, Y.K.; project administration, Y.K. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this work, the authors utilized ChatGPT (OpenAI) for language polishing and improving readability. Afterward, they reviewed and edited the content as necessary and take full responsibility for the publication’s content.

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